IN A TWO-DIMENSIONAL NOETHERIAN LOCAL RING AN ELEMENT AVOIDING THE MINIMAL PRIMES EXTENDS TO A SYSTEM OF PARAMETERS STABLE UNDER SMALL PERTURBATIONS

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Allowed standard prerequisites (editorial).

- Existence of a system of parameters extending a given element avoiding the minimal primes. invoked at proof lines 38122-38123. Supplies the element g used in the perturbation argument.
- In a Noetherian ring a power of the radical of an ideal lies in the ideal. invoked at proof lines 38124-38125. Gives the exponent n from which N = n + 1 is built.
- The relation 1 + a is a unit when a lies in the maximal ideal. used at proof line 38132. Is what makes the first generator absorb the perturbation without changing the ideal.

Source reference. Original lemma 00IM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 684–684. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 38086: sources/stacks-more-algebra/pinned-original-source.tex, source lines 38086–38086. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let $(R, \mathfrak{m})$ be a Noetherian local ring of dimension two, and let $f \in \mathfrak{m}$ be an element not contained in any minimal prime ideal of R. Then there exist $g \in \mathfrak{m}$ and $N \in \mathbf{N}$ such that*

(a) *f, g form a system of parameters for R.*

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(b) *If $h \in \mathfrak{m}^N$, then $f + h, g$ is a system of parameters and $length_R(R/(f,g)) = length_R(R/(f+h,g))$.*

Proof. By Lemma 38086 there exists a $g \in \mathfrak{m}$ such that f, g is a system of parameters for R. Then $\mathfrak{m} = \sqrt{(f,g)}$. Thus there exists an n such that $\mathfrak{m}^n \subset (f,g)$, see Algebra, Lemma 00IM. We claim that $N = n+1$ works. Namely, let $h \in \mathfrak{m}^N$. By our choice of N we can write $h = af + bg$ with $a, b \in \mathfrak{m}$. Thus

$$(f + h, g) = (f + af + bg, g) = ((1 + a)f, g) = (f, g)$$

because $1 + a$ is a unit in R. This proves the equality of lengths and the fact that $f + h, g$ is a system of parameters. □